



An explainable artificial intelligence – human collaborative model for investigating patent novelty[☆]

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ABSTRACT

With the accumulation of technology-related big data, including the patent database, existing studies have proposed a framework for patent analysis using natural language processing models. Artificial intelligence (AI) applications require human experience and insight based on the understanding of complex environments and uncertainties and model predictive performance. However, existing research has focused on applying big data and developing automated processes. Actual user understanding and the consideration of model usability are insufficient. Studies must consider the human–machine cooperation-based approach in developing the AI model. This study proposes a collaborative approach through which the explainable AI (XAI) model, a self-explaining deep neural network for text classification, communicates with users. The proposed XAI model provides users with an explanation for the model prediction along with the prediction results for patent evaluation. Users provide feedback based on the model predictions and their explanations. The source XAI model is refined via relearning by reflecting on user feedback. This study experiments to assess model improvement using the human collaboration method. As for the human collaborative method, this study considers the process of human intervention independent of the XAI model's results as well as the method of human participation based on the explanation presented by the XAI model. The experimental results verified the XAI model performance, showing the highest accuracy (0.890) and F1 score (0.916), such that the model can be applied efficiently to patent evaluation. The XAI–human collaboration model presented in this study can also be expanded and applied to technology intelligence tasks. However, the collaborative approach in this study has complete trust in human advice from technical experts; thus, subsequent collaborative XAI models could be improved by communicating bidirectionally with human resources as a complementary relationship.

1. Introduction

The technology planning cycle is shortening, and the level of research and development (R&D) is advancing.¹ In accordance with this environment, the role of technological innovation and planning is emphasized at the corporate and national levels. Technological innovation indicates the results at the level of implementation and commercialization through technology development, moving beyond a technology concept (Gans and Stern, 2003). That is, technological innovation leads to new businesses, including products, processes, and services, as a driving force that directly or indirectly affects socioeconomic growth (Yoon, 2008).

Regarding this advancement, for a patent to be registered, the novelty, inventive steps, and industrial applicability of the technical idea in the patent document must be recognized through institutional review, comprising the registration requirements set forth by the Korean Intellectual Property Office. The number of applications for intellectual property rights in South Korea reached a record 600,000 in 2021. Additionally, the number of international patent applications worldwide exceeded 3.45 million in 2022. Patent documents, extensively accumulated as big data, serve as a source of technology intelligence on various research topics where technology trend analysis, promising technology prediction, and technology road mapping flourish (Abbas et al., 2014).

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¹ https://github.com/hjzzang/FEEDBACK_XAI_GIT.

As artificial intelligence (AI) technology develops and as the patent database accumulates big data, deep learning model-based patent intelligence research has been conducted (Krestel et al., 2021). Existing studies have proposed a framework for patent analysis using natural language processing (NLP) models for various tasks, including patent summary (Trappey et al., 2020), semantic information extraction (Chen et al., 2020), patent risk prediction (Chi and Wang, 2022), and patent novelty detection (Chikkamath et al., 2020). Despite the advancement of AI technology, deep learning models have been debated because their prediction results cannot be explained (called a “black box;” (Adadi and Berrada, 2018; Castelvetti, 2016; Rai, 2020)). Patent analysis plays a vital role in providing objective evidence that supports decision-making for technology R&D planning and strategy.

However, existing AI models focus on improving predictive performance, posing difficulties in interpreting model prediction results. In technology R&D planning, the roles of various participants, including technology experts, working-level staff, and management, are essential. Establishing business and national R&D plans depends on consultation (Wang et al., 2010) and workshops (Phaal et al., 2003) with technical experts. Research outcomes, derived by data-based systematic methodology, provide decision-making materials in the field of technology management. Thus, a model must predict accurate output and explain the basis for the predicted value. However, existing research has focused on applying big data and developing automation-based processes. The understanding of the actual users and the consideration of the usability of the model are insufficient. Developing a white box of AI models so that humans can understand AI decisions is necessary to overcome the limitations of the previous studies (Hassija et al., 2024). We believe an explainable AI (XAI) model can address the challenge of black-box deep learning models. Beyond the prior XAI models, a framework in which AI models and users can interactively cooperate plays an overarching role in the field of technology management, including patent examination.

This study aims to develop an XAI-human collaborative model for patent novelty. Through the suggested collaborative approach, the XAI model can communicate with human users. A target task of the proposed model focuses on classifying the patent novelty whether a given patent is novel compared to the prior patents. The XAI model provides users with an explanation of the reason for the model prediction and the prediction results for patent evaluation. Users provide feedback to the model based on the model predictions and explanations. The source XAI model improves through relearning by reflecting on user feedback. We explore the following research questions: (1) Why is the XAI model and XAI-human collaboration necessary for technology management problems, and what are its strengths? (2) Using the explanation provided by the XAI model, how should user participation be organized, and what type of feedback should be provided? (3) How can feedback collected through user participation be reflected in the model, and what effect does it have on improving the model?

The remainder of the paper is organized as follows. Section 2 articulates the backgrounds where prior studies have reviewed text mining for patent analysis, an XAI model, and an XAI-human collaboration model. Next, Section 3 proposes a research framework to develop an XAI model for text-based patent novelty analysis, followed by the resultant dataset, model, and experiments. Then, Section 4 presents the analysis results of applying the proposed approach, and Section 5 discusses the results. Finally, Section 6 concludes the paper with the research contribution, its limitations, and further study opportunities.

2. Background

2.1. Text mining for patent analysis

Technology intelligence plays a role in establishing an R&D strategy that identifies potential opportunities and risk factors by systematically collecting and analyzing technical information necessary for decision-making (Kerr et al., 2006). As a vast volume of technical information

(e.g., patents, academic papers, and technical analysis reports) has accumulated, textual information explaining the core ideas serves as a vital source for technology intelligence (Yoon and Park, 2004). When analyzing technical information, a structural approach should be taken, not from an individual technical perspective but from a systematic perspective, that performs technical functions related to technical components at the upper and lower levels (Sun and Tan, 2012). For analysis from the technical system viewpoint, various attempts to structure information based on the key phrases, clauses, and vocabulary have been conducted. Specifically, technical functions, principles, and components can be derived from text information described as the key phrases, clauses, and vocabulary. Subject-action-object (SAO) structures (Moehrle, 2010; Yoon and Kim, 2012) comprise a representative text-mining approach to analyze technology information by extracting words or phrases that refer to the subject, object, and verb as action.

Early studies in text mining of patent analysis focused on extracting technical keywords, including “term frequency-inverse document frequency” (Ramos, 2003) and “bag of words” (Zhang et al., 2010). Yoon and Park (2005) proposed a methodology for constructing the technological morphology using a keyword extraction approach to discover technology opportunities. Developed research has discovered new technology opportunities through patent mapping using the technology keyword analysis approach (Lee et al., 2009) and exploring promising R&D fields through knowledge mapping (Yoon et al., 2010).

Additionally, research has been conducted based on topic modeling that extends beyond the keyword level to identify specific technical content. For example, Park et al. (2015) suggested a methodology to search for R&D cooperation partners by identifying patent applicants with similar R&D technical topics using latent semantic analysis. X. Wang et al. (2020b) developed a model to evaluate a company’s technological competitiveness based on deriving technological themes using the latent Dirichlet allocation model. For more specific content-based patent text mining, myriad studies are being conducted to structure technical information by specifying viewpoints, such as technical principles, solutions, and configurations.

Patent text mining has evolved from extracting text information to representing text-type data as numerical vectors to support applying them to technical management problems. Early research based on patent text vectorization created a patent map by expressing vectors with the presence or absence of keywords included in patent documents (Son et al., 2012). However, due to the data sparsity in the keyword vector-based approach, a problem emerged regarding the methodological validity. Recently, research on NLP and deep learning models has demonstrated applications in various fields of patent text mining. A study was conducted to embed the SAO structure, focusing on a function representing the core technical information (Kim et al., 2020). The suggested SAO vector can be applied to analyze patent infringement (Kim and Yoon, 2021) or explore technological opportunities (Jang et al., 2022) based on functional similarity.

Recent research has extended to additional analysis based on the embedding approach for vectorized patent text (Yu et al., 2024). Existing studies that have performed patent document embedding have primarily applied pretrained natural language models, such as word2vec (Mikolov et al., 2013) and bidirectional encoder representations from transformers (BERT) (Devlin et al., 2018). A patent document can be embedded with representation techniques to calculate the technical similarity analysis between a pair of embedded patents. A methodology has been proposed to explore technology opportunities, which calculates semantic similarities between a patent and trademark document (Lee et al., 2020). A previous study discovered commercialization opportunities based on textual similarities between technology fields derived from patents and academic papers (Takano and Kajikawa, 2019). Most existing studies have approached using embedding vectors as a part of a series of processes.

Another study investigated an approach to implementing a learning model for patent analysis by applying an embedding vector. Previous

studies have also attempted to embed patent text documents and apply them to document classification tasks (Lee and Hsiang, 2019; Li et al., 2018; Risch and Krestel, 2019; Xiao et al., 2018). The technical classification code, which is patent bibliographic information, was used and applied to the classification problem. This study also targeted a type of document classification task using a pre-trained embedding model for patent documents.

In addition to the text classification task, Hong et al. (2022) developed a model that predicts the value of an idea by applying a vectorized idea representation of the technical content as input data through patent document embedding. Jang and Yoon (2021) developed a model that classifies technical semantic relationships, including the technology class, material, causality, action, method, and physical property, in which they encoded the patent embedding and syntactic structure graph. Separately, a study proposed developing a model for summarizing patents through a generative adversarial network, with patent document embedding as input (Kim and Yoon, 2022).

Table 1 summarizes the primary methodologies and results of existing studies related to patent text mining and divides them into two perspectives: text extraction and patent text vectorization. Specifically, this study focuses on a patent novelty prediction model based on pre-trained text embedding in terms of structuring technical information. Text extraction aims to extract critical information, such as keywords and topics, according to specific rules or probabilistic statistical approaches to summarize technical content or build an ontology, which refers to the detailed concept of a targeting technology consisting of essential components (Jeong and Yoon, 2015). In contrast, patent text vectorization focuses on presenting a learning model targeting a specific task based on deep learning and NLP. Text extraction is a text mining approach that focuses on summarizing the main technical content of patent documents with a large amount of information. On the other hand, text vectorization converts the entire content of the patent document into numerical information and uses it as a source to implement algorithms such as deep learning. Both approaches have meaningful uses in patent analysis. This study analyzes patent novelty using embedding of patent text. In addition, text information that has a major influence on novelty judgment is extracted and presented as an explanation. Compared to previous studies that focused on one approach, this study has some contributions to its useful value. This study aims to determine the novelty of a patent by receiving a given patent document as input and learning it based on features that consider the technology characteristics. A text vectorization approach was applied to express the claim document, which is the core technical concept of the patent document.

2.2. Explainable artificial intelligence model

Although deep learning model accuracy continues to improve, the problem of opacity that obscures a model continues (Williams, 2021). A black-box model loses trust in using AI, which can create resistance. Moreover, AI models incur imperfections in many situations, such that the training data size and model performance cannot yield sufficient explanations for the results. For AI models that train on primarily complex hidden situations, the number of training data, amount of training, and model performance cannot produce sufficient explanations

for the validity of the results. Further, AI models require interpretation because imperfections exist in many situations (Doshi-Velez and Kim, 2017). In addition, XAI aims to provide a tangible explanation of how a model generates predictions. Specifically, XAI explains the rationale for the model decision-making process and presents process strengths and weaknesses, so that it can sense how the model would later behave (Rai, 2020).

The XAI model explanation techniques can be classified according to the method of presenting the model explanation, target task, and data type (Linardatos et al., 2020; Qian et al., 2021). Depending on the scope of model applicability, the methods can be divided into model-specific and model-agnostic approaches. Model-specific approaches work based on details about the specific model structure. These approaches leverage the internals of machine learning models and provide explanations for model results based on reverse engineering approaches.

In contrast, the methods in the model-agnostic approach do not consider the model structure. Post hoc models aim to provide local explanations for model prediction (Antoniadis et al., 2021), which can be applied to any machine learning algorithm and can work with black-box model approaches. The model-specific approaches embody layer-wise relevance propagation (Montavon et al., 2019), providing explanations based on the model parameters. The model-agnostic approach includes an approach for inductive reasoning using interpretable post hoc models, such as the local interpretable model-agnostic explanation technique (Li et al., 2024; Ribeiro et al., 2016a). Depending on the method used in the explanation model, the scope of the explanation target segregates into local and global explanations. Local explanations focus on explaining individual data instances, whereas global explanations attempt to understand the model as a whole, typically using a group of data instances to create one or more description maps (Das and Rad, 2020). Prior research has developed around model-agnostic approaches, which have the advantage of being compatible with a variety of popular libraries, whereas model-specific approaches are still in the early stages of research (Belle and Papantonis, 2021).

This study proposes an XAI model that provides local explanations related to patent novelty using a self-explaining approach. Existing studies indicate that the XAI models provide explanations through post hoc models (Zhuang et al., 2020, 2021). Teso (2019) indicated that post hoc-based explanations tend to provide unfaithful and fragile explanations, leading to user confusion. The surrogate-based explanation incurs a limitation in that it hardly determines who is responsible for the error between the learning and ex-post explanation model (Kumar et al., 2022). In other words, the surrogate approach requires understanding the learning and explanation models because the learning and explanation of the model operate independently.

Notably, this study aims to apply the XAI model to the technological management field, a complicated and complex R&D environment. Actual users of the model proposed in this study are focused on using the results of the deep learning model, and limitations exist in understanding the structure and operation of the model. In addition, the approach must include additional procedures for practical interpretation and application after the model prediction, burdening the working level, including patent examiners responsible for analyzing the novelty of patents and R&D organizations at companies and research institutes. Therefore, this study aims to increase the usability of the XAI model

Table 1
Prior studies on text-based patent mining.

Approaches	Main methods	Research outcomes
Text extraction	- Keyword extraction - Topic modeling - Structuring technical information	- Summarizing technical content - Technology ontology construction
Text vectorization	- Vector representation (keyword vector, SAO2vec) - Pretrained text embedding models (Word2vec, BERT, etc.)	- Calculating technical similarity approach - Classification/prediction model for technology information - Summarization model for technical documentation

based on a self-explaining method, where the model result can be applied directly without an additional surrogate approach to draw an explanation.

The XAI-based models have been applied in various NLP fields, including text classification (Liu et al., 2018; Mathews, 2019), question answering (Ghosh et al., 2019; Thayaparan et al., 2020), and sentiment analysis (Zucco et al., 2018). Danilevsky et al. (2020) classified the existing studies on XAI-based NLP models into (1) description via inputting the data feature importance, (2) surrogate models for explanation, (3) example-oriented input explanations based on similar instances, (4) provenance-based explanations, and (5) declarative induction. Revina et al. (2021) presented an XAI model pipeline for text classification. Danilevsky et al. (2020) suggested that the challenge of XAI research in the NLP field lies in the generation, display, and comprehension of explanation methods from users' perspectives across the entire cycle. This study focuses on developing a classification model for patent novelty evaluation. We aim to design an XAI model that can emphasize the usability for the examiners or R&D practitioners.

2.3. XAI-human collaboration model

Over time, XAI models have been evolving into a form of human-machine collaboration using model explanations (Hartmann and Sonntag, 2022). The general framework of the human collaborative XAI model comprises the following three processes (Fig. 1): (1) providing explanations, (2) offering feedback, and (3) updating the model. First, when the model provides explanations, the XAI model presents its scope and technique. Second, human feedback involves settings and feedback forms of experiment participants. Human participants in experiments are chosen by specific selection, crowdsourcing, or simulation. Additionally, human feedback can correct answer labels and input words related or unrelated to correct the answers and word importance scores. Finally, a model improvement plan can be designed by revising model parameters, learning the dataset, and learning the process.

The present study aims to propose an XAI-human collaboration model based on the explanation-based human debugging framework in Fig. 1. The XAI model provides predictions along with explanations to users. Users, including the selected participants (Kulesza et al., 2010) and crowdsourced participants (Smith Renner et al., 2020), can provide feedback on the results, facilitating the identification and correction of errors in the model and leading to model updates.

Moreover, XAI-human collaboration research is expressed based on various keywords, such as a debugging model that emphasizes model error correction, human interaction, human-in-the-loop, and human-centric features. This study reflects how the XAI model and humans interact. Based on this research background, we reconstructed previous studies in Table 2. The structure considers how to organize the participants, the kind of feedback provided to obtain the XAI model results, and how those results were reflected in the model.

Previous studies were reviewed for examples that present the model design and implementation results rather than the conceptual or

Table 2

Studies of the explainable artificial intelligence (XAI)-human collaboration model.

Participants	Feedback forms	Model update	References
Nonstakeholder (research students)	Model prediction and word relevance	Dataset augmentation	Kulesza et al. (2010)
Nonstakeholder (research students)	Word relevance	Revised model restrictions	Stumpf et al. (2009)
Nonstakeholder (research students)	Model prediction	Dataset augmentation	Teso and Kersting (2019)
Nonstakeholder (cloud sourcing)	Model prediction and word relevance	None (expectations for model improvement)	Smith Renner et al. (2020)
Nonstakeholder (cloud sourcing)	Input feature relevance	Feature removal	Lertvittayakumjorn et al. (2020)
Nonstakeholder	Input feature relevance	Revised regularizer in the loss function	Yao et al. (2021)
Simulation-based data labeling	Model prediction	Dataset removal (relevant to misclassification)	Koh and Liang (2017)
Simulation-based data labeling	Model prediction	Revised data labeling	Khanna et al. (2019)
Simulation-based data labeling	Model prediction	Revised data labeling	Han and Ghosh (2021)
System designers and end users	Word relevance	Irrelevant word removal (not relevant)	Ribeiro et al. (2016b)
One expert user	Similarity between cases	Dataset augmentation	Zylberajch et al. (2021)

theoretical work. Considering the participant composition, most existing studies have consisted of student participants who were not related to the model. Other studies have employed a simulation approach in which data labels are manipulated without direct human participants. For example, Ribeiro et al. (2016b) proved the effect of model improvement by organizing participants with basic knowledge of the topic (religion) and a learning dataset field, as opposed to nonexpert participants. However, the classification task for religious relevance accommodates a field that does not rely heavily on expertise compared with medicine (Hartmann and Sonntag, 2022; Rong et al., 2022; Schaekermann et al., 2020). In addition, Dikmen and Burns (2022) emphasized the role of experts with domain knowledge in XAI use. Further, Zylberajch et al. (2021) involved one expert user participant, omitting that expert's field and purpose in the study.

In summary, existing studies have not sufficiently considered stakeholders, including model planners, developers, actual users, and regulators, for the XAI model in participant composition. Additionally, the nature of the model may vary depending on whether the user has

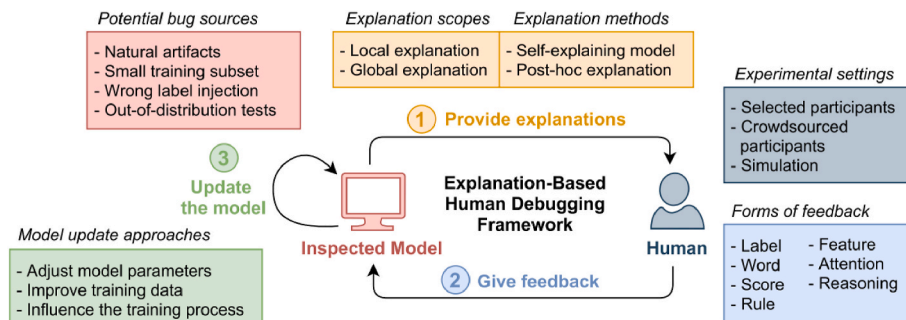


Fig. 1. General framework of the explanation-based human debugging framework (Lertvittayakumjorn and Toni, 2021).

expert knowledge about the XAI model (Langer et al., 2021).

However, prior conceptual studies have proved that stakeholder consideration is crucial in successful AI-human collaboration (Ehsan et al., 2021; Langer et al., 2021; D. Wang et al., 2020a). Kotnis et al. (2022) presented a guide for human-centric research in the NLP field based on the importance of stakeholder consideration, referencing domain experts, data collectors, researchers, developers, and end users in a series of research stages that include problem definition, data selection, model design, and task evaluation (Kotnis et al., 2022).

In previous studies, participants have provided feedback on relevance at the level of classification labels, words, and input features, which are all predicted values of the model. In the method of reflecting model feedback, an approach was performed to augment the training dataset (Kulesza et al., 2010; Teso and Kersting, 2019; Zylberajch et al., 2021) or modify the label (Han and Ghosh, 2021; Khanna et al., 2019). That is, previous studies have approached the method of correcting the input dataset, yielding considerable influence on model misclassification. This approach does not conform to the XAI-human collaboration interaction, a two-way process in which models receive feedback from humans and improve themselves to match human opinions. For example, Yao et al. (2021) applied the relevance of the input feature as feedback at the word level, adding it as a regularizer in the model loss function, which calculates the error between the correct answer and predicted values when training a deep learning model, also known as the cost function. The model received additional input from human feedback, using the input for learning.

In addition, Chromik and Butz (2021) derived the following seven types of human interaction with XAI based on the scope and goal of the interaction: transmission, dialogue, control, experience, optimal behavior, tool use, and embodied action. Based on the descriptions and according to the type they presented, this study aims to apply XAI interaction in the control approach. The approach moves beyond XAI interaction for information transmission by updating the model behavior to the level required by the user based on user feedback. We focus on the model users, organizing the participants and reflecting their feedback in the model loss function to improve the model.

3. Research framework

3.1. Research concept

Fig. 2 presents the conceptual diagram of the proposed XAI-human collaborative patent evaluation model. Initially, the XAI model for patent novelty emerged as the source model. The results of the source XAI model are provided to the users, and accordingly, XAI model retraining for refinement is performed based on their feedback.

The XAI-human collaborative patent evaluation model consists of three modules (Fig. 3): dataset construction, XAI-human collaboration model design, and human collaboration experiments. In the first module, a binary label dataset for patent novelty and a user advice dataset for the target patent are constructed. A technology expert patent examiner derived the advice dataset by reconstructing a search report. The search report includes the classification label for the novelty evaluation result and the claim number of the related prior patent when it is determined that novelty is infringed. The second module of the proposed XAI-human collaboration model is performed by learning a source model and a refinement model, reflecting the user advice. In the final module of experiments, we considered the timing of this advice regarding whether it was reflected by the model explanations and target fields of technology.

3.2. Dataset

The XAI-human collaboration model dataset for patent evaluation is divided into a source model and an advice-based refinement model. Both datasets are structured by processing the European Patent (EP) full-text dataset (<https://www.epo.org/en/searching-for-patents/data/bulk-data-sets/data>) provided by the EP Office (EPO). The dataset includes a search report written by a technical expert examiner, as presented in Fig. 4. The source model classifies the patent novelty between the target and prior patents. We designed a training dataset including labels and advice with 1) binary labels using the relevance to novelty infringement and 2) the related claim index information tagged by the examiner in the search report. The advice dataset, which is input only into the refinement model, plays a role in revising the claim weight derived from the source model. In the example, Fig. 4 of the search report indicates that target patent EP 10016203 is determined to infringe upon the novelty (Category X) compared to a prior patent, EP 0832752A2. It further indicates that the target patent Claims 1–5, 7–10, and 12 are highly relevant to the portion from column 15 on line 22 to column 19 on line 47 of the prior patent. The proposed model constructs the input using the word token information from the claim documents of the target and prior patents. The user feedback comprises binary information, specifically the ID information, regarding the relevant claims of the target patent related to the infringement of novelty in the prior patent.

This study proposes a method to label the patent novelty by considering the successive examination history, as listed in Table 3. Novelty labels are constructed for pairwise patents according to cases in the patent history of the target patent. The final unit of input data is the pairwise patents between the target and prior patents that the target patent cited. In the proposed human-XAI collaboration model, the user advice was formed by restructuring the European search report, which is a report on the investigation of the potential infringement of prior patents written by EPO examiners, into a dataset. The last column of Table 3 indicates whether the advice dataset can be constructed depending on whether the EPO search report is included. The patent history includes applications, revisions, registrations, withdrawal, and rejection, where EPO patent identification informs the history in the first two digits (A1, A9, and B1 refer to the European patent application published in the search report, the corrected A document, and the granted patent, respectively; <https://www.epo.org/en/searching-for-patents/helpful-resources/first-time-here/definitions>). The data labels, which include novel or novelty-prejudicial, are classified into three cases based on the

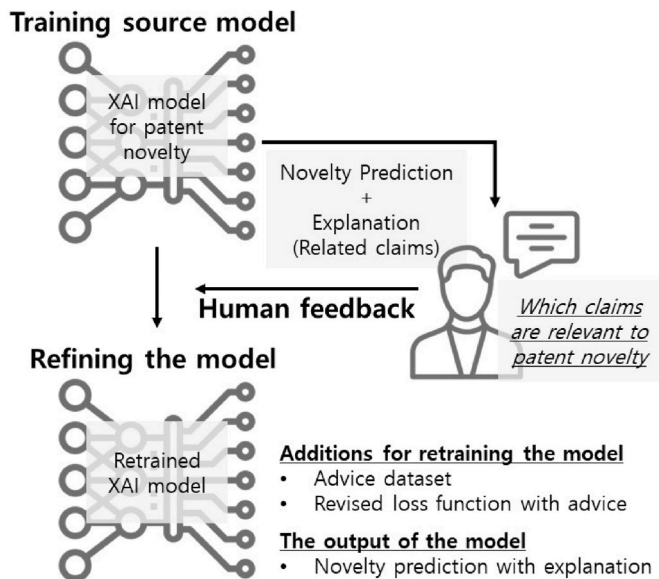


Fig. 2. Conceptual diagram of the explainable artificial intelligence (XAI)-human collaborative model for patent novelty.

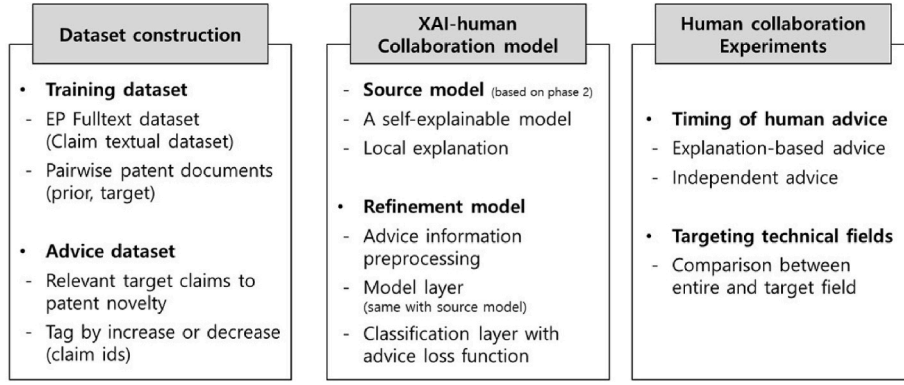


Fig. 3. Research framework.

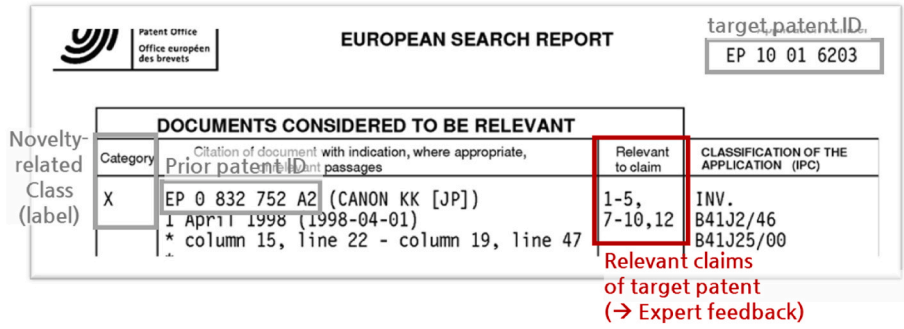


Fig. 4. Example of a search report from the European Patent Office (EPO).

Table 3

Data labeling through patent history.

Cases of patent history		Pairwise patents (target – cited)	Novelty label (binary classes)	Human advice (included or not)
Case 1	A1 → B1	A1(B1) claims – cited claims	1 (Novel)	Included
		Selected A1 claims – cited claims	0 (Novelty-prejudicial)	Not included
Case 2	A1 → A9 → B1	A1 claims – cited claims	0 (Novelty-prejudicial)	Included
		B1(A9) claims – cited claims	1 (Novel)	Not included
		Selected A1 claims – cited claims	0 (Novelty-prejudicial)	Not included
Case 3	A1 → withdrawal/refusal	A1 claims – cited claims	0 (Novelty-prejudicial)	Included
		Selected A1 claims – cited claims	0 (Novelty-prejudicial)	Not included

evaluated patent. Case 1 is recognized as a novel patent in which the patent subject to evaluation is registered as it was filed. Case 2 is changed into a novel patent in which an application is registered after amendment. The last case is not identified as a novel patent, where an application was filed but withdrawn or rejected after the examination. The XAI model training dataset considers the document unit, including the partial claim set related to the novelty judgment, and the entire claim.

3.3. XAI-human collaboration model for patent novelty

The XAI-collaborative model for patent evaluation comprises a pro-

cess of first learning the source model and then learning an additional improvement model that reflects the technical expert advice. The model architecture includes preprocessing, input, interpretation, and output layers for the input patent document pair. The composition of the XAI model in this study follows the model architecture presented by Sun et al. (2020) and Jang et al. (2023). The source model receives the claim documents of the target and prior patents and learns a classification task to determine whether the target patent is novel, as illustrated in Fig. 5 (left). After learning the source model, the model receives additional user advice and retrain the revised model, as depicted in Fig. 5 (right). The advice is concatenated with the original input vector as a binary vector according to the relevance of the novelty judgment in the claim unit of the patent. The encoded user advice readjusts α , which is the importance of the claim pairs in the model.

Notation for the model explanation is as follows.

- An input claim document (target patent, prior patent):
 - target $cl_{ti} = \{cl_{t1}, cl_{t2}, \dots, cl_{tN}\}$ (N : length of claim in the target patent)
 - prior $cl_{pj} = \{cl_{p1}, cl_{p2}, \dots, cl_{pM}\}$ (M : length of claim in the prior patent)
- Input advice:
 - $Adv_k = \begin{cases} 1 & \text{(if an expert advises claim}_{ti} \text{ is novelty – prejudicial)} \\ 0 & \text{(else)} \end{cases}$ (K : length of $claim_{ti}$ in the target patent, $k \in K$)
- Claim document representation:
 - $W \in \mathbb{R}^{V \times D}$ (V : vocabulary size)
- Predicting label y for x based on $p(y|x)$:
 - Input cl : associated with label $Y \in \mathcal{Y}$
 - Source model loss function $\mathcal{L}_{src}$
 - Refinement model loss function $\mathcal{L}_{rft}$

Preprocessing treats the textual technology information in the target and preceding patent claims. The input layer of the source model forms

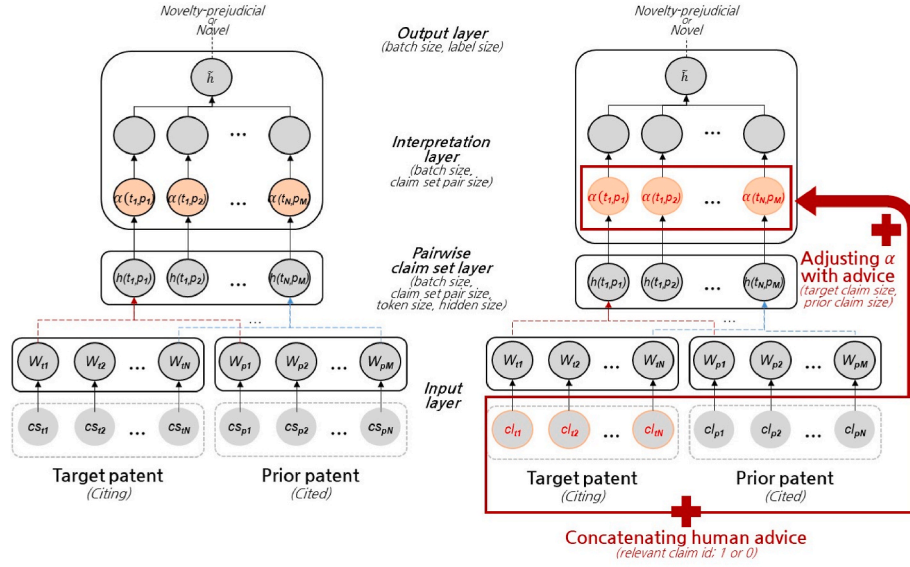


Fig. 5. Model architecture. Left: source model, Right: human-XAI collaboration model.

the batch data, including the embedded vector (target cl_{ti} , prior cl_{pj}) and novelty label (label Y). In this study, the input layer develops at the level of a pair of claims between the target and prior patent documents. The input layer is constructed by aggregating the word tokens within the claims of the target patent and the corresponding prior patent document pair. This aggregation process forms the input data for the model. The embedding vector for the claim is encoded using the pretrained Roberta model (Liu et al., 2019), which is a prominent pretrained embedding model extensively applied in patent document embeddings (Bekamiri et al., 2022; Haghighian Roudsari et al., 2022). We set the maximum number of tokens to 512, the maximum tokens of the Roberta model (https://huggingface.co/docs/transformers/model_doc/roberta), and the number of claims to 33, which is the maximum number of claims without outliers, in consideration of the computing resource size. The input layer of the refinement model adds the advice dataset aggregated to the original tensor. Advice information appears as binary information on related claims reviewed by the technical experts. The advice information size is identical to the maximum number of claims for the source model.

The proposed interpretation layer in this study uses the locally self-explainable approach. Primarily, the information on the claim of the evaluation target and the preceding patent received at the input layer arrives at the hidden layer node $h(cl_{ti}, cl_{pj})$ using the feedforward network (FFN) as a mapping function. The neural network learns sequentially from the input layer through the hidden layer to the output layer (Svozil et al., 1997). The hidden unit of $\tilde{h}$ is learned by adding $\alpha(cl_{ti}, cl_{pj})$ corresponding to the relevance score for the novelty judgment of the evaluation patent.

This study advocates a parameter of self-explainability α to learn the relevance score of a pair of claims between target and prior patents as a regularizer. The value of α is trained on the model to analyze the relevance of the related claims that are the basis for determining the novelty. The last output layer calculates the probability distribution of the label by applying the softmax function, where all input values are normalized to values between 0 and 1 as output, and the total output value is 1. The self-explainable model is learned by reflecting a regularizer α in the loss function. The loss function of the source model is calculated by combining the probability distribution for the label with a regularizer α adjusted by λ . In other words, λ is a parameter that controls the size of influence of α , which represents the relevance reflecting the claim id pair and human advice for it in the interpretation layer. The loss function of the refinement model reflects the advice Adv_i provided by a technical

expert. Concurrently, if claim i relates to the lack of novelty, Adv_i equals zero and vice versa. The method of adjusting the weight of the hidden layer through advice references the attention mechanism (Vaswani et al., 2017). Proving that the attention mechanism in the FFN can solve the problem in a long text input (Raffel and Ellis, 2015) demonstrates that the mapping function of the suggested interpretation layer is suitable. A description of each layer follows.

- (1) The input layer is expressed as a D -dimensional embedding vector through the pretrained RoBERTa method for the target patent claim cl_{ti} and the prior patent claim cl_{pj} .
 - Target patent claim documents $\mathcal{W} [cl_{ti}, :]$
 - Prior patent claim documents $\mathcal{W} [cl_{pj}, :]$
- (2) The pairwise claim set layer is located at the top input layer comprising k encoder stacks, and the p th node is derived by $h_p^k \in \mathbb{R}^{(len \text{ of } cl_p \times D)}$.
- (3) The interpretation layer derives $h(cl_{ti}, cl_{pj})$ and $\alpha(cl_{ti}, cl_{pj})$ by calculating the relevance score of the saliency (α) through a combination of patent claims.

$$- H(cl_{ti}, cl_{pj}) = F(h_{ti}^k, h_{pj}^k), ti \in \{1, N\}, pj \in \{1, M\}$$

* Applying the FFN as a mapping function F

$$- \alpha(cl_{ti}, cl_{pj}) = \frac{\exp(\tilde{h} \cdot h(cl_{ti}, cl_{pj}))}{\sum_{ti \in \{1, N\}, pj \in \{1, M\}} \exp(\tilde{h} \cdot h(cl_{ti}, cl_{pj}))}$$

$$- \tilde{h} = \sum_{ti \in \{1, N\}, pj \in \{1, M\}} \alpha(cl_{ti}, cl_{pj}) \cdot h(cl_{ti}, cl_{pj})$$

- (4) The output layer applies the softmax function to the loss functions of each source and refined model:

$$- \mathcal{L}_{src} = \log p(y|x) + \lambda \sum_{ti} \alpha^2(t_i, p_i)$$

$$- \mathcal{L}_{rfm} = \log p(y|x) + \lambda \sum_{ti} (\alpha(t_i, p_j) * 2 * Adv_i)^2$$

$$\left(\sum_{ti, pj} \alpha(t_i, p_j) = 1, \lambda \sum_{ti} (\alpha(t_i, p_j) * 2 * Adv_i)^2 = 1 \right)$$

* Lambda (λ) is a parameter of the self-explainability to adjust the total effect of the relevance score.

3.4. Experimental framework

This section presents the design of an experiment for an XAI-human collaborative patent evaluation model, focusing on improving its utilization when the XAI model and technology experts cooperate in technology planning and management. After learning the source model, the refined model reflecting technical expert advice is relearned. The results include a comparison between the performance of the refined model with that of the source model. In addition, user feedback can be influenced by the explanations in the XAI model.

Bond et al. (2019) proposed the need for XAI and studied the bias that occurs when users overly trust and accept computer-based advice. Therefore, this study designed an additional experiment in consideration of user bias for XAI. The additional target of this module is to compare the results of human providing feedback on only selected claims that are highly relevant to the novelty judgment presented by the XAI model (Fig. 6) with the results of a human independently providing feedback on all claims (Fig. 7). The additional experiment verifies whether the collaboration method based on the explanation presented by XAI can demonstrate the effect of selection and concentration. We consider the timing and method of providing the technical expert's advice. Patent novelty is analyzed as a unit of claim in which the claim ID is a kind of patent bibliometric information given according to the number of claims.

The proposed XAI model provides an explanation of the related claims, and a user produces advice regarding whether or not the claims presented by the XAI model are related, as displayed in Fig. 6. The relevant claims that the XAI model judges to be relevant for prediction are provided as an explanation; thus, users can refer to the explanation from the model. However, because a human offers feedback based on these explanations, the individuals may exhibit a bias toward a given explanation. We convened an experiment where humans can confirm relevant claims that exceed the pre-set threshold as an XAI explanation. Afterward, we only included advice from technical experts that satisfies the conditions above the threshold. In this experimental setting, we reflect on the role of XAI for humans. The users of the XAI model can specify the threshold. For example, the threshold can be set as a condition above a specific value, and the top n cases can be selected. When the threshold is set to 0.2 in Fig. 6, as an example, values for Claims 1 and 5 are higher than the threshold and are provided to the technical

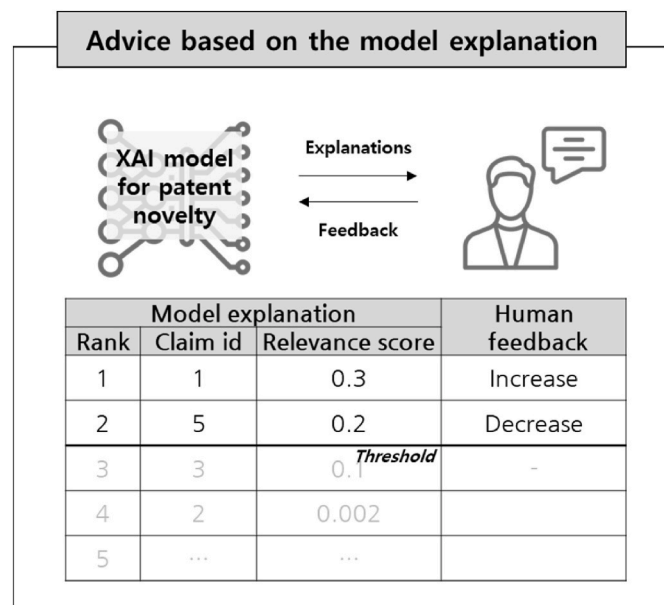


Fig. 6. Experiment A: Advice based on the explainable artificial intelligence (XAI) results.

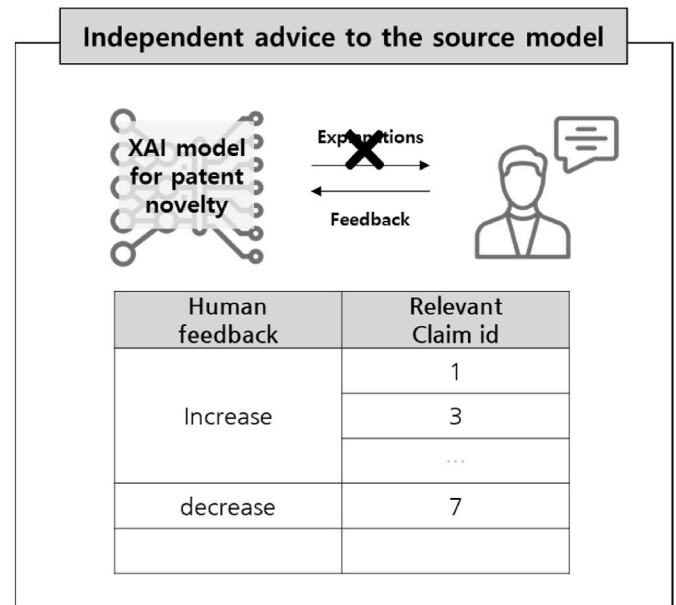


Fig. 7. Experiment B: Independent advice in the explainable artificial intelligence (XAI) results.

expert. The user provides feedback by determining whether Claims 1 and 5 are relevant to determining novelty.

In another experiment, a controlled environment emerges such that a human independently advises the XAI model without being provided an accompanying explanation. As displayed in Fig. 7, the person verifies all claims included in the evaluated patent and provides advice on claims that must be increased or lowered in importance to determine the novelty. In Fig. 7, the refinement model is learned by increasing the model parameter importance for Claims 1 and 3, which are judged to be of high importance when validated by the technical expert, and decreasing the opposite case. In this way, the technical expert does not refer to the explanation from the XAI model, and the refined model relearns with more emphasis on the feedback provided independently by the expert. In this case, the role and utilization of the XAI model may be limited, and the degree of user intervention may increase. However, this approach may be free from this influence if the XAI model provides an incorrect explanation.

In constructing human advice information, this study processed the EP full-text dataset provided by the existing EPO instead of experimenting with direct user participation. Advice data were developed for cases where an examiner presented the same opinion in the search report provided by the EP full-text dataset for the top three related claims suggested by the XAI model. The XAI-based framework for assessing novelty in patents is proposed for the first time in this study. Therefore, criteria, such as specific thresholds or rankings based on relevance, should be established depending on the intended application. This research expects to use the proposed model to provide decision support for those involved in patent examination and comparative analysis. This study assesses user feedback for the three claims with the highest relevance score values by learning the XAI source model.

4. Results

4.1. Dataset

In this study, the source model and improved model datasets were constructed with the goal of developing a collaborative model between XAI and humans for evaluating patent novelty. The dataset was constructed as conveyed in Table 4 by labeling according to the three cases of patent history proposed in Section 2. The learning dataset for the

Table 4
Dataset.

Patent history	Pairwise patents	Novelty label	Number of data
Case 1	A1 claims – prior (cited) claims	1 (Novel)	7969 (including advice)
	B1 claims – prior (cited) claims	1 (Novel)	7969
	Selected A1 claims – prior (cited) claims	0 (Novelty-prejudicial)	7969
Case 2	A1 claims – prior (cited) claims	0 (Novelty-prejudicial)	39 (including advice)
	B1 claims – prior (cited) claims	1 (Novel)	39
	Selected A1 claims – prior (cited) claims	0 (Novelty-prejudicial)	39
Case 3	A1 claims – prior (cited) claims	0 (Novelty-prejudicial)	3447 (including advice)
	Selected A1 claims – prior (cited) claims	0 (Novelty-prejudicial)	3447

source model consisted of 25,069, 117, and 6894 cases for each overall case, resulting in 32,080 cases. Case 1, in which the patent specification (B1) was submitted as a patent application publication (A1), accounted for the highest portion, with 7969 and 8550 cases, respectively.

However, even after the publication of the application, the data were expanded to improve the learning performance for cases in which the specification was written in a different technical vocabulary. The advice dataset was expanded by separately configuring “selected claim information” related to patent infringement in the search report written by the examiner. The search report was attached only to the applicant patent (A1); thus, the training dataset of the improvement model was constructed for the patent application publication (A1) document to which the search report was attached for the patent history. The final number of data according to the patent history type were 7,969, 39, and 3,447, respectively. The total number of datasets according to each case was 30,918, of which 15,977 cases were labeled “Novel (1)” and 14,941 cases were labeled “Novelty-prejudicial (0).” The total cases were divided into training, validation, and testing datasets with a 6:2:2 ratio to learn the source and base model. User advice included 11,455 cases (37 %) of the total dataset. The dataset with advice was divided into equal proportions and was used to learn the refined model after learning the source model.

The learning data examples were derived from the processing of the EP full-text dataset provided by the EPO. The data from the source model include text pairs of target and prior patent claims with novelty labels. The text pair data appear in the form of an aggregated list in which each claim text is embedded by the pretrained model (Liu et al., 2019). The training dataset of the refinement model includes advice data along with claim text pairs and label information from the source model. The advice data field consists of a list for which the length is the number of claims to be evaluated, defined as 1 for a claim number related to the novelty examination and 0 otherwise. For example, in Case 1 in Table 5, all claims from 1 to 15 are related to those of the prior patents in evaluating their novelty. The advice dataset was represented by 1 for all 15 list elements to mark the novelty-related information. In Case 2, we built an advice dataset with 14 claims where Claims 12 and 13 were interpreted as being related to the lack of novelty of the prior patent. Specifically, this study includes relationship advice information for the lack of novelty at the claim level, as the most critical element in evaluating patent novelty.

4.2. Model results

In this section, the XAI-human collaboration model examines the implementation results of the XAI-based source and refined models, reflecting the technical expert advice. Experimental XAI results and the

Table 5
Example of data cases.

Patent history	Pairwise patents	Novelty label	Advice data
Case 1	[[‘An article of footwear, comprising: ...’, ‘The article of footwear of claim 1, ...’ ...], [‘A midsole assembly for an athletic shoe comprising: ...’, ‘A midsole assembly for an athletic shoe ...’ ...]]	1	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
Case 2	[[‘A process for the prediction ...’, ‘The process according to claim 1, ...’, ...], [‘A bioreactor for the cultivation ...’, ‘The bioreactor according to claim 1, ...’ ...]]	1	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0]
Case 3	[[‘A method for manufacture of food products, ...’, ‘The method of claim 1, ...’, ...], [‘A reduced calorie cocoa butter composition, ...’, ‘The reduced calorie cocoa butter composition according to claim 1, ...’ ...]]	0	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0]

interpersonal, collaborative model for patent novelty evaluation are discussed. A model classifying a lack or the recognition of patent novelty was tested. We employed accuracy and the F1 score as performance evaluation indicators. The F1 Score is the harmonic mean of precision and recall. Precision is calculated by dividing the number of true positive results by the total number of samples predicted to be positive, including incorrect predictions. Recall is calculated by dividing the number of true positive results by the total number of samples that should have been identified as positive. For the additional interpretation of the F1 score, the precision and recall values were presented together. All evaluation indicators were derived as macro averages of the indicator values for each label class. An imbalance existed in the number of cases per label in the study data; thus, the average was derived after calculating the index value according to the label rather than evaluating all cases in a wide area.

In this study, an experiment was conducted by adjusting the lambda corresponding to the effect on self-explainability. The lambda value was adjusted up to 1, considering the value exhibiting the model performance improvement, including 0 without the effect of self-explainability. The experimental setting of the lambda value follows the experimental settings from Sun et al. (2020). The dataset was split at a respective ratio of 6:2:2 for training, validation, and testing. Table 6 lists the experimental settings of the XAI learning model. For the word embeddings, we adopted the hyperparameters specified in the RoBERTa model (Liu et al., 2019). The experimental design was based on the layer size settings employed in Jang et al. (2023). Table 7 presents the accuracy and F1 score according to the model experiment. The source model results were derived by learning the XAI model 20 times without technical expert advice. After source model learning, the refined model was trained 20 additional times by receiving input from the technical expert’s advice dataset. Thus, the source and refined models were trained for 20 iterations. The number of iterations of the experiment was set by considering the convergence point for the model performance and the

Table 6
Experimental settings.

Experiment items	Experimental settings
Word embedding	- RoBERTa model (default setting)
Layer size	- Token size: 512 tokens (256 tokens each) - Hidden size of interpretation layer: 768
Hyperparameter	- Learning rate: 2e-8, lambda: 1.0 (default) - Optimizer: AdamW, max epoch: 20
Model performance	- Evaluation metrics: loss, accuracy, F1 score - Monitor with valid accuracy (mode = max)
Computing resources	- GeForce RTX 3090 24 GB

Table 7

Model performance.

λ	Accuracy			F1 score		
	Source model	Model update		Source model	Model update	
		Base model	Refined model		Base model	Refined model
0.0	0.631	0.842	0.853	0.585	0.870	0.881
0.6	0.589	0.849	0.890	0.652	0.879	0.916
0.8	0.613	0.771	0.773	0.676	0.784	0.787
1.0	0.531	0.588	0.594	0.578	0.558	0.549

internal computing resources.

The suggested model of XAI-human collaborative approach, marked as the refined model in the table above, performed better than the source model. The results indicate that the base XAI model can be improved by reflecting the user advice, specifically the technical expertise in this study. The suggested XAI-human collaborative model achieved the highest accuracy (0.890) and F1 score (0.916) when lambda was 0.6, whereas the base model accuracy was derived as 0.589, an improvement of 0.301. The base model was trained 20 additional times after the source model without using the advice to form a base model. The suggested human collaborative XAI algorithm showed improvement by 0.041 compared to the base model, which is an identical approach to Jang et al. (2023). The experimental results reveal that the performance of the base model improved after an additional 20 training sessions following the initial 20 sessions of training on the source model. Furthermore, the base model, trained for a total of 40 sessions, exhibited stable performance, suggesting that the model proposed in this study requires a minimum of 40 iterations.

We compared the performance of the base model, which underwent additional learning by applying a model identical to the source model, and the refined model, which relearned the model by reflecting the user feedback. In other words, the difference in learning performance between the base and refined models can verify the effect of improving the model performance through user feedback. In this study, user feedback is based on the search results of patent examiners who are technical experts in a specific domain. The refined model, developed using the feedback data, performed better than the base model. The performance improvement effect of the refined model derived from Table 8 indicates that the information on critical claims provided by technical experts was significantly helpful in predicting novelty.

The experimental results of the model for the lambda (λ) value that controls the self-explainable effect (Section 3.3) are as follows. The experimental results of the refined model revealed that the F1 score increased from 0.549 to 0.916 according to the lambda value. Thus, the performance of the source model according to lambda was the highest at 0.631 when lambda was 0 (i.e., not self-explainable). Thus, a higher lambda value results in lower performance. The experimental results for the source model demonstrated a trade-off relationship between the model explanatory power and performance, as indicated in previous studies. However, when the model was improved with the advice of the expert, as proposed in this study, the self-explainable approach yielded performance superior to the source model, overcoming the limitations of the existing self-explainable approach.

Table 8

Model performance of advice types.

λ	Accuracy		F1 score		Precision		Recall	
	Type 1	Type 2	Type 1	Type 2	Type 1	Type 2	Type 1	Type 2
0.6	0.890	0.888	0.916	0.915	0.931	0.929	0.890	0.888
0.8	0.773	0.769	0.787	0.783	0.826	0.822	0.773	0.769
1.0	0.594	0.588	0.549	0.558	0.608	0.604	0.594	0.588

4.3. Experimental results

In the XAI-human cooperative model, the improvement effect of the XAI model may vary depending on how a user presents advice. This section compares the performance of the refined XAI model, considering the approach used to refer to the XAI model results when providing technical expert advice. The technical expert advice was based on the search report, including the claim information related to the prior patent novelty infringement and novelty inhibition, which the examiner prepared. This information was written by the technical experts, apart from the XAI model, during the actual patent examination process.

Accordingly, we designed the following two experiments. Type 1 represents a method in which user feedback is provided independently in the XAI model as a baseline. Type 2 involves a method in which a user verifies the prediction results and explanations of the XAI model and presents feedback. Among the prediction classes and explanations presented by the XAI source model, the advice dataset was constructed only for claim information related to the relevance score of the top three patent pairs (evaluation target and prior patents). That is, this study verifies the improvement effect of the XAI model when the examiner examines all claims and provides advice on the critical claims suggested by the XAI model.

Table 8 lists the results of the model experiments according to the human advice method. The Type 1 model accuracy and F1 score values were higher for all lambda values compared with the values for Type 2. The experimental setting that produced the optimal performance occurred when lambda was 0.6, where the accuracy was 0.890 and 0.888 for Types 1 and 2, respectively. The Type 1 and 2 F1 scores were 0.916 and 0.915, respectively. Regardless of the presentation of the XAI model results, the XAI improvement effect of the expert advice through the examination of all Type 1 patent claims was higher than that of the top three presented Type 2 claims. However, the difference in the F1 value was insignificant at 0.001. Thus, providing feedback for all claims more significantly affects performance improvement, given a greater absolute volume of provided information. However, the difference in performance improvement between the two types occurred at the level of 0.001 (standard when lambda is 0.6); thus, significant improvement can be expected with a more efficient user feedback presentation method than simply the maximum volume of information. The experimental results can be interpreted as reference material for the results presented by the XAI model.

4.4. Applications

This chapter introduces the practical application of the framework proposed in this study. In this study, we developed a model to evaluate the technological novelty of the target patent by analyzing pairs of target patents and prior patents. In particular, the XAI model presented in this study is a method of relearning the base model with human cooperation. The proposed model explains not only the binary class for the novelty evaluation results of the target patent, but also the basis for the novelty judgment in terms of related patent claims. An example case presented by the inference model is shown in Table 9. The target patent "EP2608599A1" is a patent for Internet Protocol fragmentation applied by Acer Incorporated. The prior patent US20090303947A1 is a technology for a mobile communication system applied by Fujitsu Ltd. The proposed inference model recognizes the technological novelty of the

Table 9
Example of the result of XAI for the patent novelty.

Patent pair	Patent novelty
Target patent: EP2608599A1 (Method of performing Internet Protocol fragmentation and related wireless network system)	[Claim 1] A method of performing Internet Protocol fragmentation in a wireless network system, comprising: performing data transmission between a transmitting end and a receiving end of the wireless network using a communication protocol; characterized by before transmitting a plurality of data packets, detecting a maximum transmission unit of the communication protocol at the receiving end and transmitting a corresponding synchronization signal from the receiving end to the transmitting end; modifying a cutting rule according to the synchronization signal at the transmitting end; segmenting the plurality of data packets into a plurality of sub data packets according to the cutting rule; and transmitting the plurality of sub data packets from the transmitting end to the receiving end.
Prior patent: US20090303947A1 (Mobile communication system, mobile communication method, and communication apparatus)	[Claim 2] The mobile communication system according to claim 1, further comprising a repeater including a repeater-side memory which stores the MTU value of a destination-side transmission path which is a transmission path leading to the destination of the data from the mobile station, a repeater-side acquirer which acquires the MTU value of the destination-side transmission path, and a repeater-side notifier which notifies the base station-side acquirer of the MTU value stored in the repeater-side memory, wherein the base station-side acquirer acquires the MTU value notified by the repeater-side notifier and stores the acquired MTU value in the base station-side memory.
Novelty class: 0, Explanations: claim 1 – claim 2 (0.9715)	

target patent. Additionally, the model predicted that claim 1 of the target patent hindered the novelty of claim 2 of the prior patent at a level of relevance score of 0.9715.

Actors from various perspectives can utilize the framework proposed in this study as a paractical application. First, patent examiners perform prior art searches as their main task. Patent examiners are responsible for searching prior patents in related fields and investigating whether the patent subject to evaluation satisfies the patent requirements. Technology companies can also utilize the proposed framework. Let's say a company is internally evaluating the R&D results for the technology and preparing to apply for a patent. A prior art search must be conducted to determine whether the R&D result has secured technical novelty, which is one of the review criteria for patent registration. Conversely, a company that has registered a prior patent can use it to continuously track and manage whether subsequent patents infringe on its own patents in order to protect the company's internal technology. When preparing to develop new technology, R&D planners are interested in areas of interest. You can look at related technology trends by creating a technical overview and comparing it with prior art. From these various perspectives, using the vast accumulation of prior patents, open source big data, and the framework proposed in this study can intelligently and efficiently replace large-scale time-consuming and cost-consuming tasks, including those involving technical experts. In addition, the XAI model that cooperates with humans proposed in this study can be upgraded to a model more suitable for a specific technology field by collecting additional advice from humans if you do not agree

with the evaluation results presented by the model.

The framework presented in this study signifies an endeavor to exceed the boundaries of traditional machine learning and deep learning-based patent analysis research. [Krestel et al. \(2021\)](#) conducted an extensive review of previous studies, exploring various deep learning algorithms such as classification, search, and prediction utilizing patent big data. Recently, there have been efforts to introduce XAI models capable of overcoming the inherent limitations of black box deep learning models. [Shajalal et al. \(2023\)](#) introduced a model for predicting the classification code of patents, which is particularly straightforward for application in deep learning-based patent analysis. Previous research in this series proposed a patent novelty classification model using explainable AI (XAI) ([Jang et al., 2023](#)). However, this study extends beyond the XAI model by introducing a collaborative framework where humans and XAI work together. Although combining human interaction with XAI may seem straightforward, effective collaboration involves challenges—such as developing structured protocols for human input, aligning AI explanations with diverse human perspectives, and managing cognitive biases that may affect human interpretation ([Bertrand et al., 2022](#)). In diverse R&D activities, patent analysis supports communication between experts and management, focusing on utility rather than full automation. This XAI-human collaboration aims to address these complexities, contributing to practical applications by enhancing human-AI synergy in patent analysis.

5. Discussion

Today, AI-based R&D planning can serve as an effective decision-making tool through interaction between humans and AI models. This section discusses methods to collaborate between AI (especially XAI) and humans, and explores the research implications from this perspective. [Chromik and Butz \(2021\)](#) introduced the categorized concept of interaction between XAI and humans focusing specifically on the goals and effects. The XAI model overcame the limitations of previous deep learning black-box models with additional explanations for its predicted values.

In accordance with the existing framework, this study performed a transmission-type interaction that provides an objective explanation for AI behavior. An interactive XAI model conducted research, including the explanation style it conveys to users. In this study, an additional module allows examination of the explanation types the model can provide, depending on prospective user viewpoints, such as examiners and R&D planners, and prediction classes, such as the lack or existence of novelty. Additionally, we improved the XAI source model by reflecting the advice after AI presented the predicted values and explanations to the humans. This approach can be interpreted as an interactive model of the “control type” that aims to converge the AI behavior according to the user requirements. The role of the experts’ experience and know-how in specific technology domains plays a significant role in R&D management, including patent examination ([Ernst, 2003](#)). Although human advice can have favorable and unfavorable consequences, this study focuses on the role of a credible adviser.

Further, the XAI model can be improved by communicating bidirectionally with human resources as an enhanced level of interaction category. The XAI and human can communicate with each other to advance their strengths and compensate for each other's weaknesses. While XAI models have large-scale learning capabilities, they lack understanding of the macroscopic environment, but experienced users have insight despite any limitations in memory for individual cases.

The deep learning model receives human feedback as a control signal and adjusts the model. This method should proceed through the process of exploration and verification, accounting for AI expertise and the scope of explanation (local or global). Previous studies have suggested an additional visual analysis system that derives actionable insight into how deep learning models improve ([Krause et al., 2016](#)). This study proposes focusing on the transmission-type XAI model as the most

intuitive interaction and the control type based on user requirements. This study was reflected in the model improvement with unconditional trust (i.e., without review of human judgment). A risk exists when the advice presented by the user is incorrect. Therefore, further study becomes necessary, specifically of the XAI model developed from the proposed transmission and control type XAI interactions for the embodied action type. The concept builds on theories of computer-supported cooperative work (Wang et al., 2020a) community, such as mutual goal understanding, proactive task co-management, and shared progress tracking. The interaction of the embodied action type envisions symbiosis led by the AI model and the human, rather than a controlled interaction that updates the model only according to the user requirements. In future research, a more evolved form of the XAI model can be designed by expanding it to an embodied behavioral interaction in which the AI model and humans can communicate and update bidirectionally in real time regarding a common goal.

Additionally, further research reflecting the interaction concept of the “tool use type” can be conducted. Patent evaluation and technology novelty analyses, set as the target tasks in this study, must be adjusted according to the target technology field. Patent claims may include domain-specific terms and knowledge. Depending on the technical field, the technology form, such as method, material, and structure, also develops differently. For example, when modeling is performed by selecting a specific technology field for a dataset, the performance of the refined model may vary. The proposed XAI model and dataset used in the XAI–human collaborative model were trained on all patents filed with the EPO. Therefore, the proposed model learns novelty by receiving patent data for an entire technical field. In this study, the size of the training dataset was maximized without limiting the technical field to improve model performance.

However, patent analysis can be performed by companies and research institutes that focus on a specific technology field. Patent examiners also examine patents in specialized fields based on their expertise in a specific technical field. In the practical application stage of this patent analysis, experiments can be designed in consideration of the technical field of the analysis target. Thus, when a technical expert provides advice on a target technology field, an experiment could be performed to determine whether a customized model would improve the predictions on the target field. Integrating human involvement in our automated system, although potentially increasing the time of operations, significantly enhances accuracy and reliability, especially in complex scenarios where automated algorithms may falter. While this integration may slow down the system compared to fully automated alternatives, the benefits of improved accuracy, adaptability, and compliance justify this trade-off.

6. Conclusion

This study proposed a human–machine collaborative patent evaluation model to strengthen the interaction between XAI and users. The collaboration-based framework focuses on the role of humans as technology experts and the strength of AI. The suggested model sequentially operates by training a source XAI model and refining the source model by reflecting the human advice. Accordingly, the XAI model was improved as a refined model, and this study conducted experiments to test the model improvement effect according to the human cooperation method. Regarding that method, we considered the process of human intervention with and without the model explanation. The experimental results verified the performance of the XAI model, which can be applied efficiently to patent evaluation based on the main explanations of the XAI model. The results also substantiated a novel approach to describe an applicational framework in the field of R&D management beyond the existing XAI model. This study discusses the proposed model in view of the level of human interaction and how the collaboration approach could be advanced.

The presented XAI–human cooperative novelty evaluation model has

the following academic and practical contributions. We first developed an XAI model that can interact with users in the form of information transmission. In this study, technology experts/examiners represented users in consideration of the technology novelty evaluation task. Based on the expected users of the model, advice from technical experts improved the AI model. This study is the first to present a collaborative model between XAI and users that considers the actual model users and simultaneously updates the model. The practicality and effectiveness of the research results offer a customized model according to the purpose. The role of quantitative and qualitative evaluations represents a substantial problem in patent evaluation and judgment of technological novelty. The presented XAI–human collaboration model can be employed as an analysis tool for patent evaluation and is expandable and applicable to technology intelligence tasks. The patent novelty evaluation applied in this study can be expanded to applications in various fields, including patent examiners, patent applicants, and technology management practitioners. Each user has a variety of purposes, including prior art review, auxiliary materials for claim preparation, technology planning, technology infringement response, and technology trend research. When designing a customized model, the usability and effectiveness of the model can be improved by considering the purpose of the stakeholders/users.

However, this study has the following limitations. The model improves performance only for detailed model design matters, such as the input dataset structure, parameters, and layer configuration of the XAI model. Recently, powerful and efficient models have been developed rapidly in the NLP and deep learning fields. In this study, the pretrained RoBERTa and FFN approaches were used. The latest high-performance model must be reflected when using the proposed model. Additionally, the presented XAI–human cooperation model updates the model based on the assumption that the user advice is unconditionally correct. In contrast, if a technical expert presents incorrect advice, the model could be updated detrimentally. Therefore, future research must consider enabling interaction that does not compromise opinions between the XAI model and humans.

CRediT authorship contribution statement

Hyejin Jang: Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Byungun Yoon:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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